Hall Ticket Number:

MR-21

Code: B413CS5

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular Examinations, Aug/Sep-2023

Computer Organization and Architecture

(Computer Science and Engineering)

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Difference between Computer Organization and Computer Architecture?	2	2	2	1
1. b)	Explain various types of Shift operations?	2	1	2	2
1. c)	Explain RTL statements?	2	2	2	1
1. d)	Outline the need of data manipulation instructions?	2	2	1	1
1. e)	Find the largest and smallest values that could be stored in an 8 bit signed register & unsigned register?	2	4	5	1
1. f)	What is meant by normalization in floating point representation? Why do we need it?	2	3	5	3
1. g)	What are the various I/O commands used in Input/output interface?	2	1	2	1
1. h)	How to calculate the hit-ratio in cache memory organization?	2	4	3	2
1. i)	Differentiate between RISC and CISC architectures.	2	1	4	5
1. j)	Illustrate the terms Multiprocessors and Multi Computers?	2	2	4	5

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Design a circuit transferring data from a 4 bit register uses multiplexer's to another register? And use selection circuit to select any of four 4-bit registers content as to bus. Give full explanation?	10	2	2	2
	OR				
2. c)	With neat flow chart, explain the phases of instruction cycle in digital computer.	5	2	1	1
2. d)	Design a combinational circuit to addition, subtraction, increment and decrement for 4 bit register using full adders and explain control logic?	5	2	2	3
3. a)	How do we classify CPU's instructions based on their register organizations? In which Organizations zero address instructions are used? Mention few zero address Instructions And their actual execution in practice?	10	4	1	4
	OR				
3. c)	Explain the role of an addressing mode in an instruction operand fetching? Explain various addressing modes with suitable flow diagrams?	10	2	2	6
4. a)	Given Booth's algorithm to multiply two binary numbers, explain the working of the algorithm with neat flowchart? Perform Multiply 10111 with 10011?	10	5	5	2
	OR				
4. c)	Construct the flowchart for Restore division operation for Signed numbers?	4	2	5	1
4. d)	Solve a restore division operation between 25 as Dividend and 6 as a Divisor? And show the remainder and quotient in their respective registers?	6	3	5	3
5. a)	Discuss about cache memory and how its supports to enhance the performance of a computer?	5	2	3	1

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5. b)	Given the main memory access time is 1200ns and cache access time is 100ns. The average memory access time is not to exceed 120 ns. Determine the hit ratio for the cache?	5	4	3	2
	OR				
5. c)	Compare and Contrast between Memory mapped I/O and isolated I/O	4	4	3	1
5. d)	Discuss the design of typical input/output interface?	6	2	2	1
6. a)	Explain inter-process arbitration? Give various arbitration techniques with neat sketch and explain in detail.	6	2	2	3
6. b)	Write about Time shared common bus and multiport memory.	4	1	3	9
	OR				
6. c)	Explain four segment pipelines with neat sketch and show the pipeline flow of execution through space time diagram.	5	3	4	10
6. d)	What are the various conflicts raised in the pipeline processing and give various solutions to handle conflicts.	5	2	4	12

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome